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B.TECH, C.E., MAJOR PROJECT - PHASE - II

AN APPROACH TOWARDS BUILDING A SMART LIFT SYSTEM USING MACHINE LEARNING

PROJECT GUIDE
DR. SUDHIR DHAGE

RECAP...

- Problem Focus: Energy optimization using ML based elevators
- Experimented with Single-lift systems to optimize wait times and energy consumption
- Data Synthesis: Using periodic functions
- Reductions in time as well as energy were observed on trivial data

PROJECT OBJECTIVES

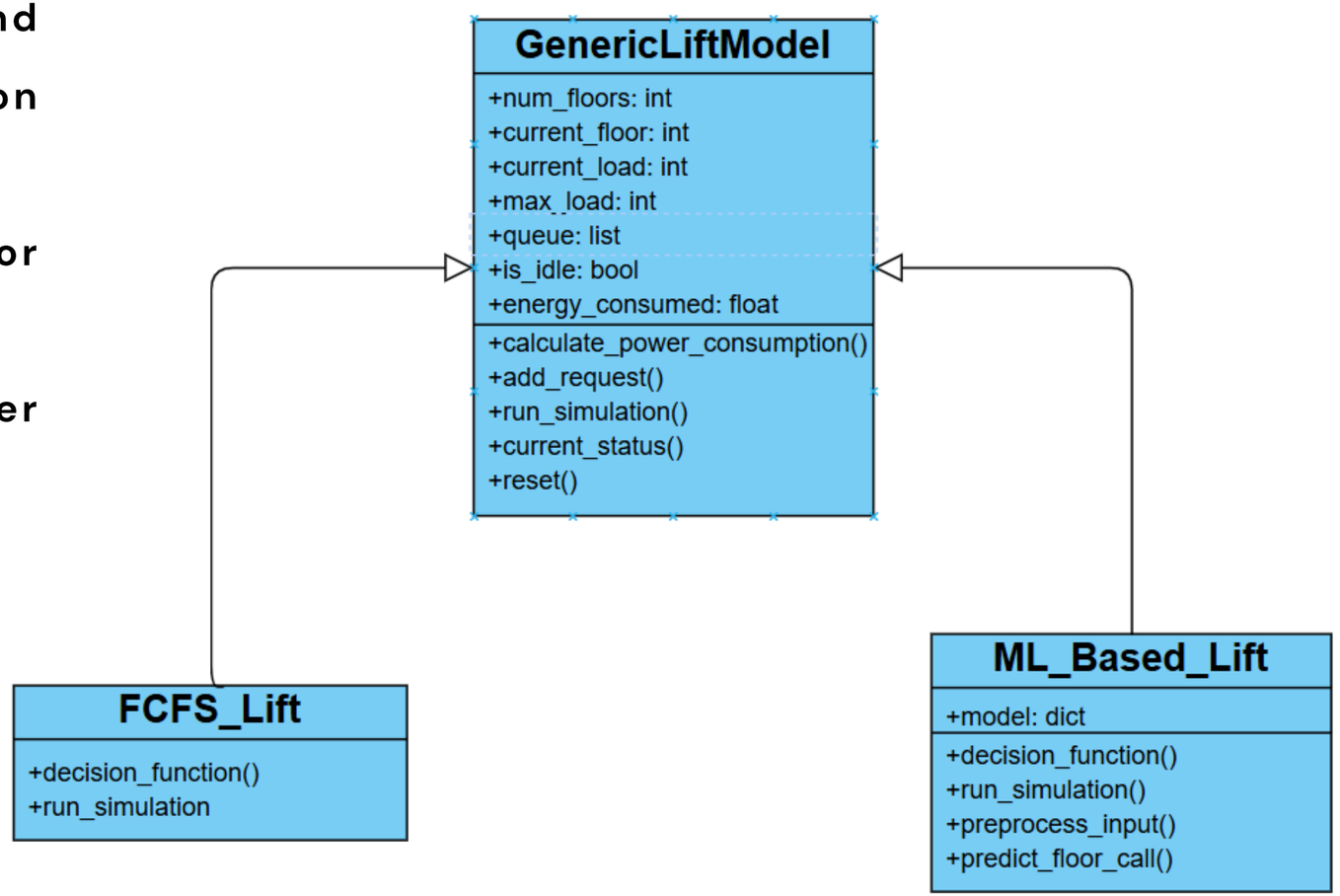
- To design lift systems that optimize power consumption and reduce overall energy usage.
- To predict lift usage patterns using machine learning for efficient routing and operations.
- To develop a scalable system adaptable to different building sizes and usage scenarios.
- To reduce unnecessary lift stops and improve the overall efficiency of lift movement.
- To integrate data-driven decision-making for smarter, real-time lift operations.

TO BE DISCUSSED...

- Simulation software progress
- LSTMs in Multi-lift system
- General interaction model
- Proposed ML models' architecture
- Results
- Conclusion
- Further work
- References

SIMULATION SOFTWARE PROGRESS...

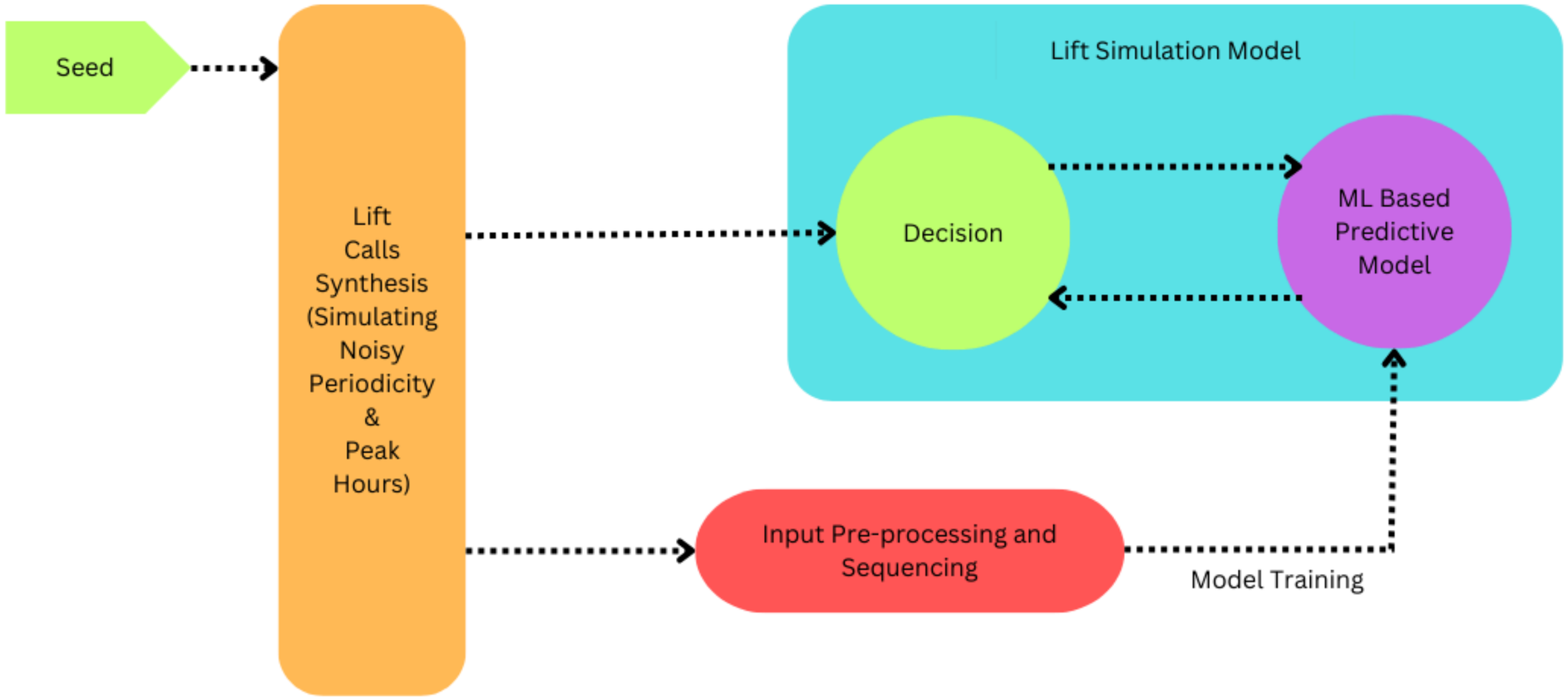
- Have developed Generic and Specific lift models using python classes
- Modularization of the tasks for research speed-up
- Tested and confirmed the proper working of each module



LSTMS IN MULTI-LIFT SYSTEM

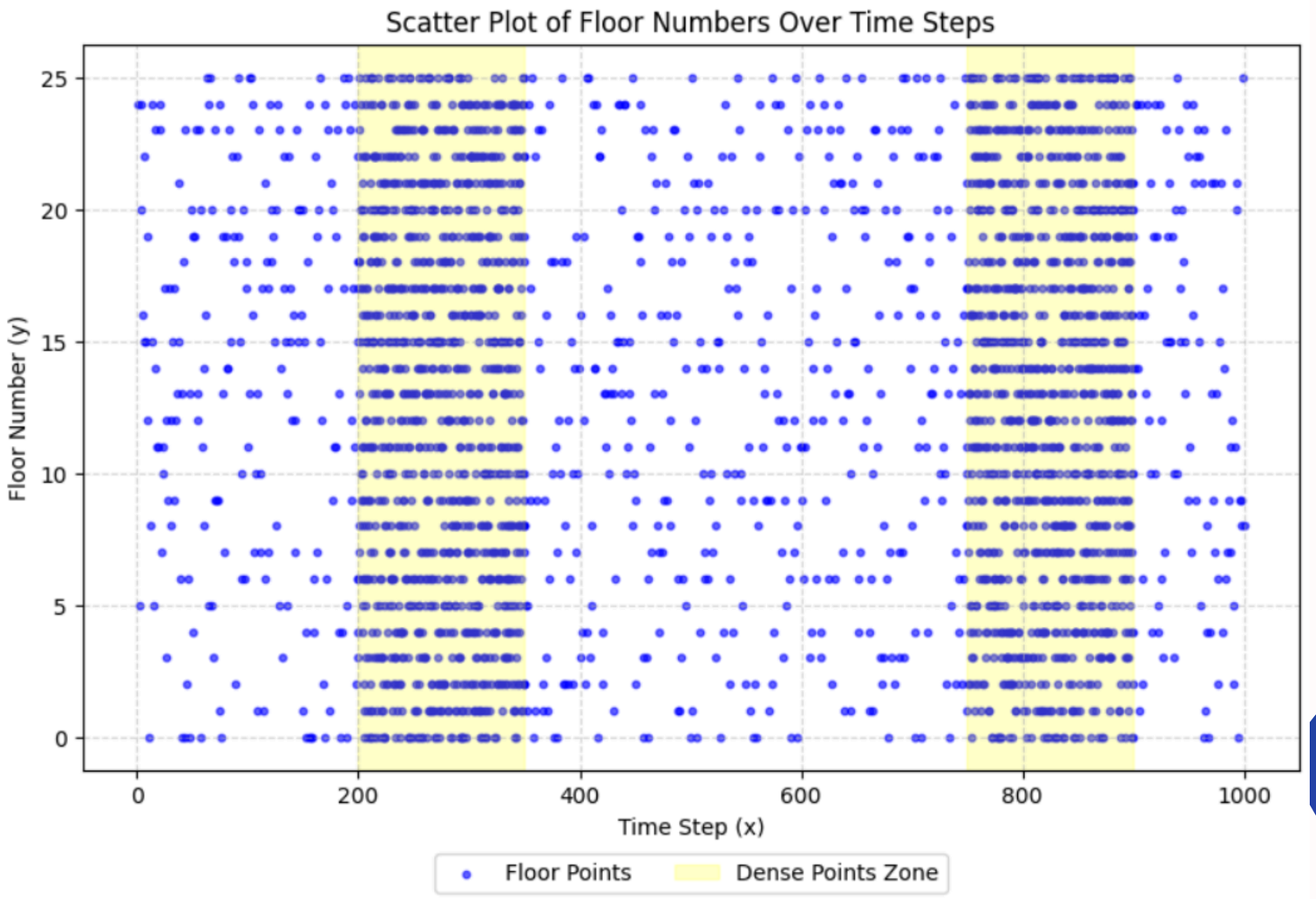
- Predict Peak Usage: LSTM models predict high-demand periods, optimizing lift availability and reducing energy waste.
- Dynamic Lift Scheduling: Predictive models ensure only necessary lifts operate, conserving energy during off-peak times.
- Optimize Load Distribution: LSTM forecasts help balance lift traffic, preventing energy waste from overloading.
- Energy-Efficient Routing: LSTM predicts the most likely next stops, minimizing unnecessary travel and saving energy.

GENERAL INTERACTION MODEL

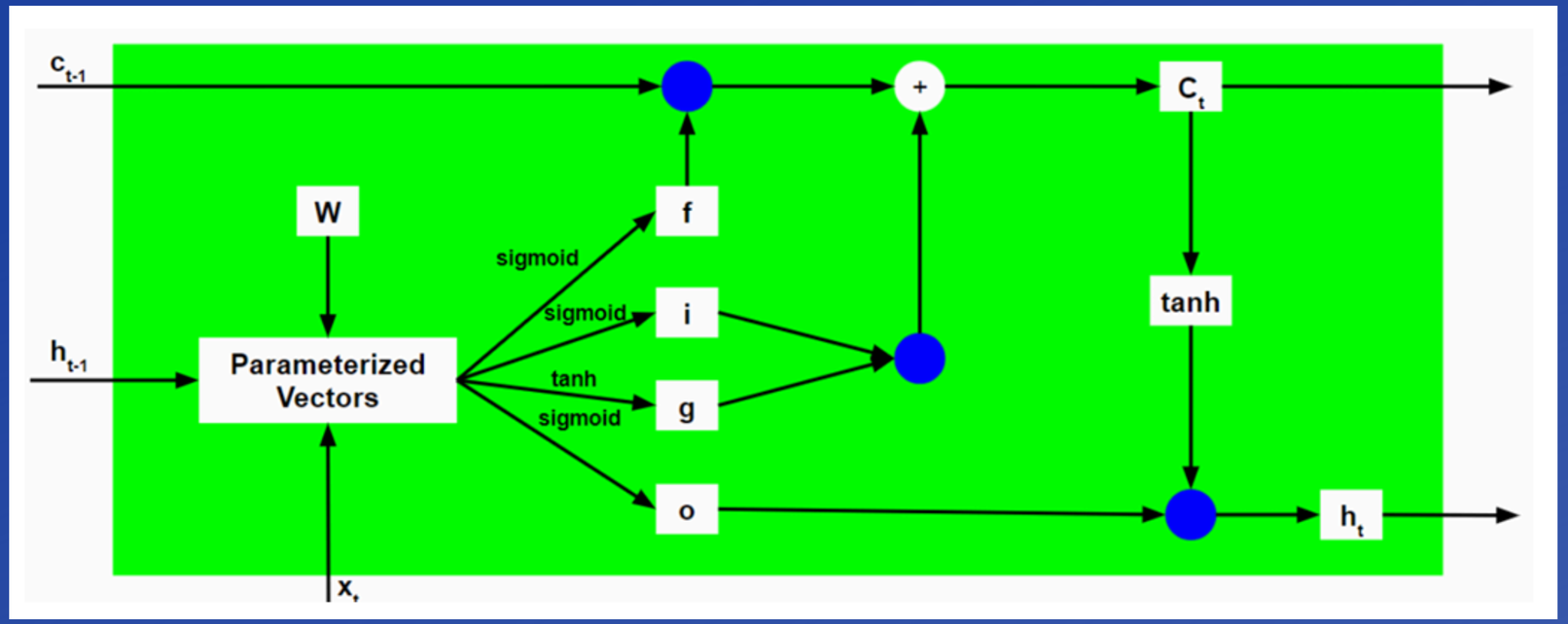


DATA SYNTHESIS TO REFLECT PEAK HOURS

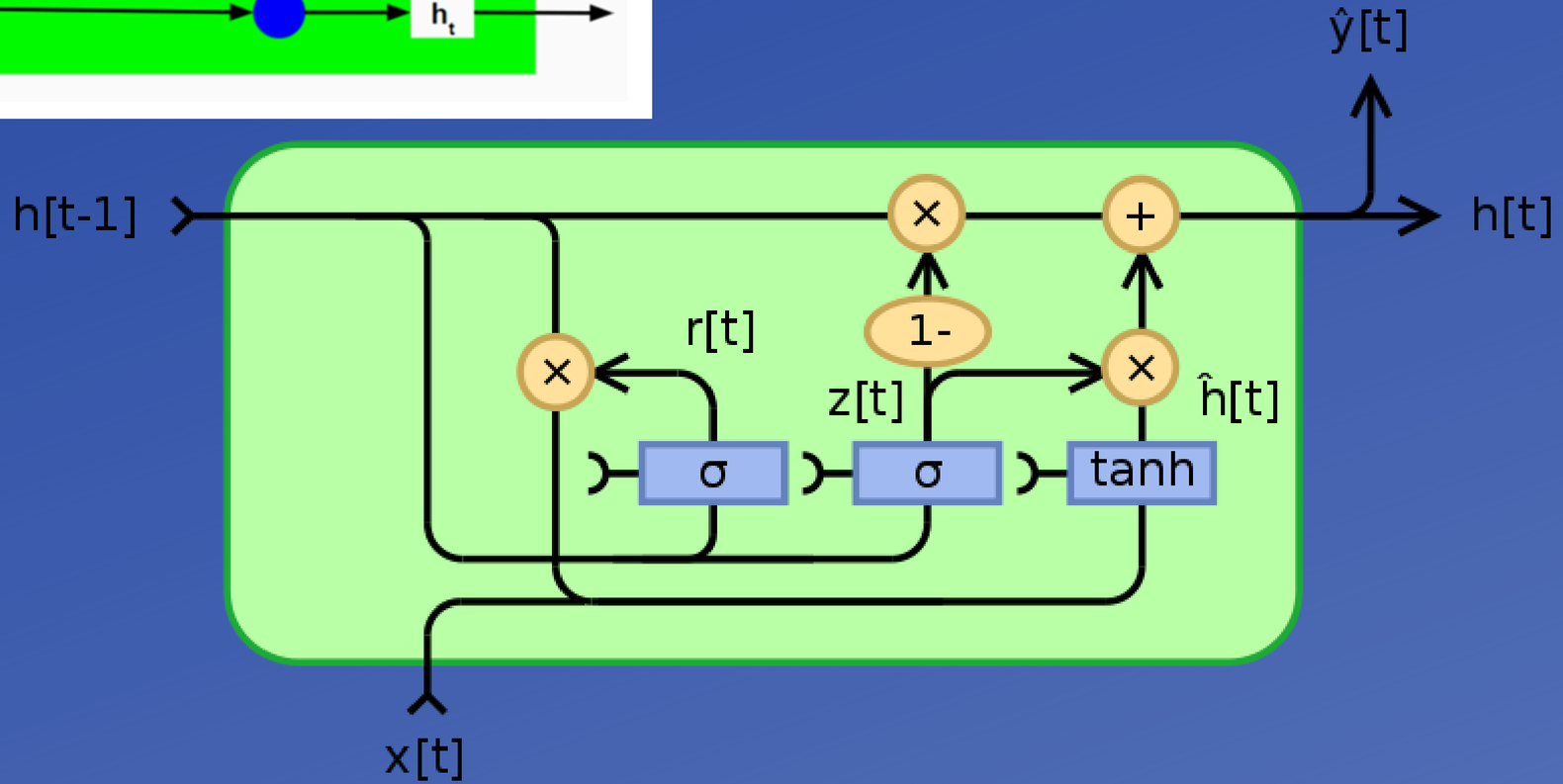
- Followed a probabilistic approach to generate data
- Peak hours decided beforehand
- Multiple random selections at peak call ranges, while single selection at non-peak ranges



PROPOSED MACHINE LEARNING MODEL ARCHITECTURES

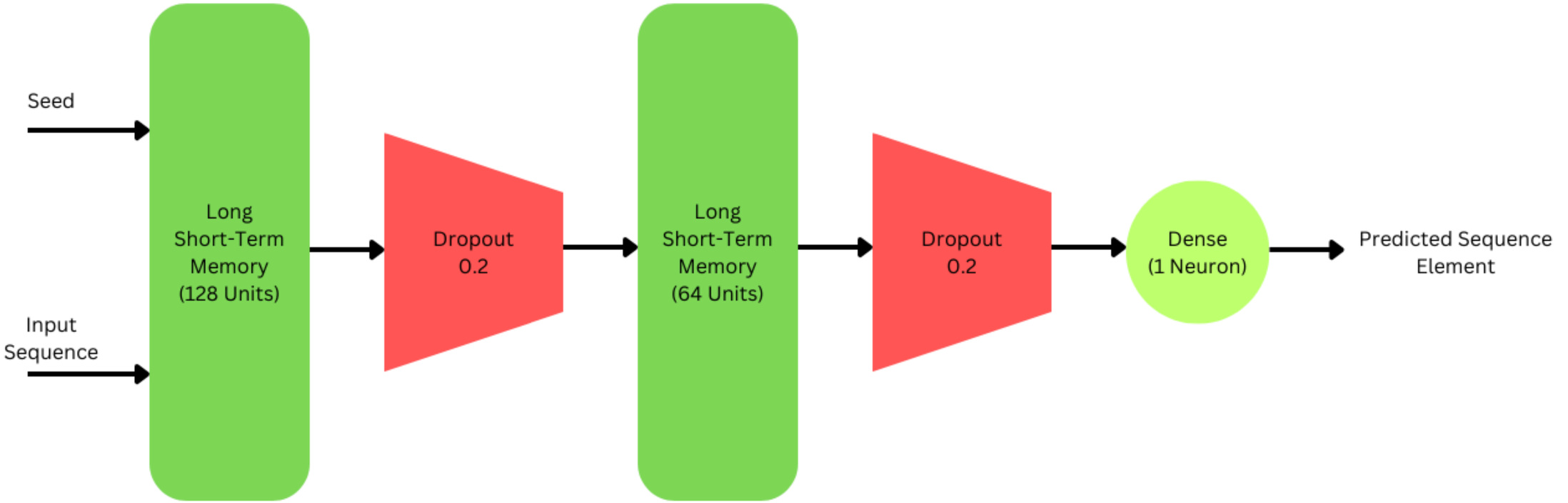


A SINGLE LSTM UNIT

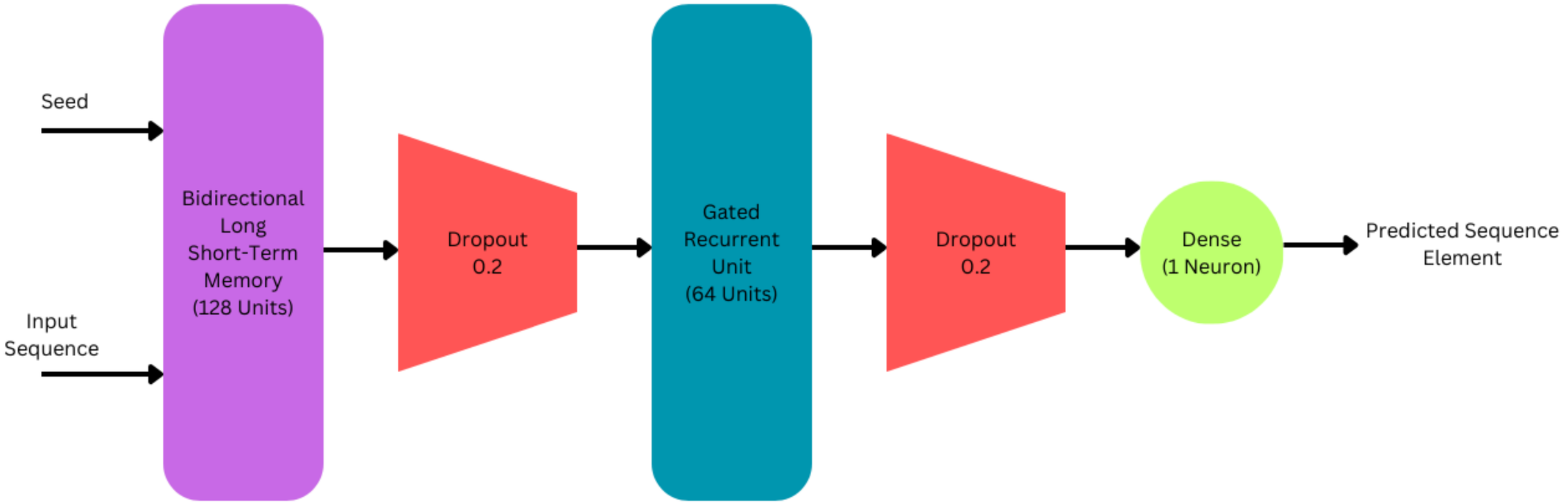


A SINGLE GRU UNIT

PROPOSED MACHINE LEARNING MODELS
1. PURE LSTM ARCHITECTURE

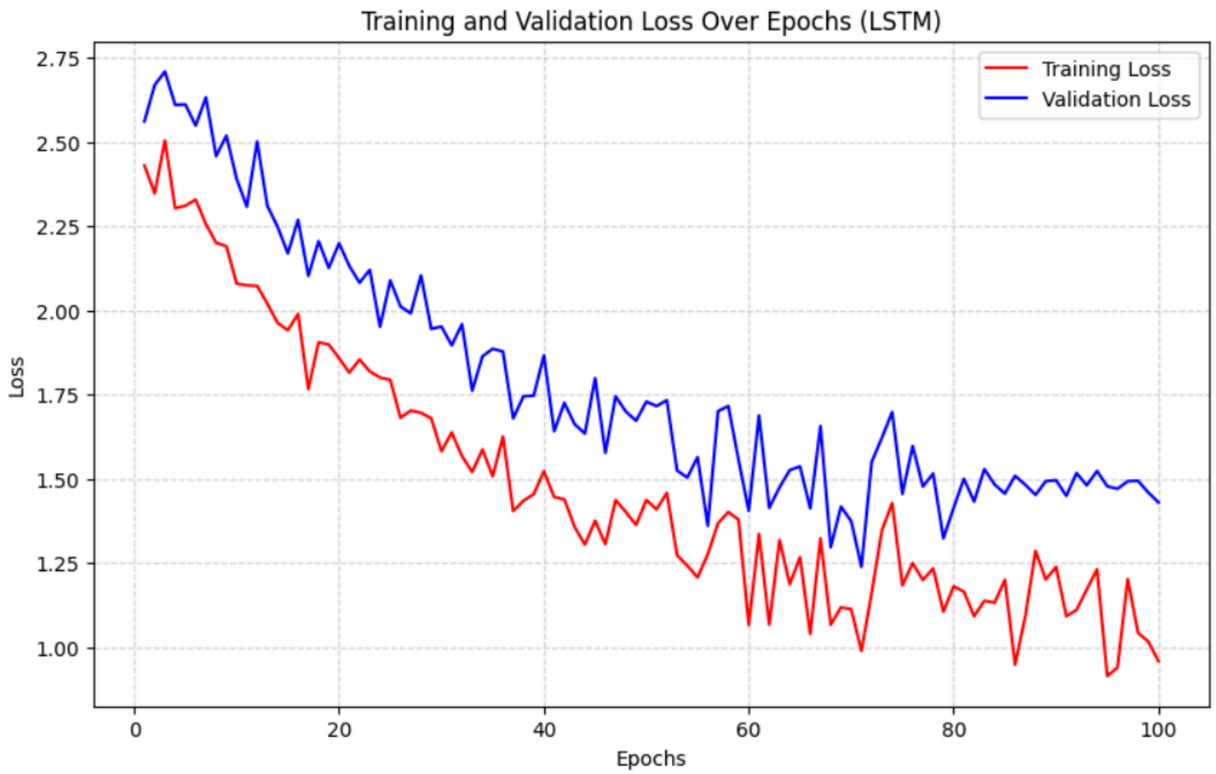


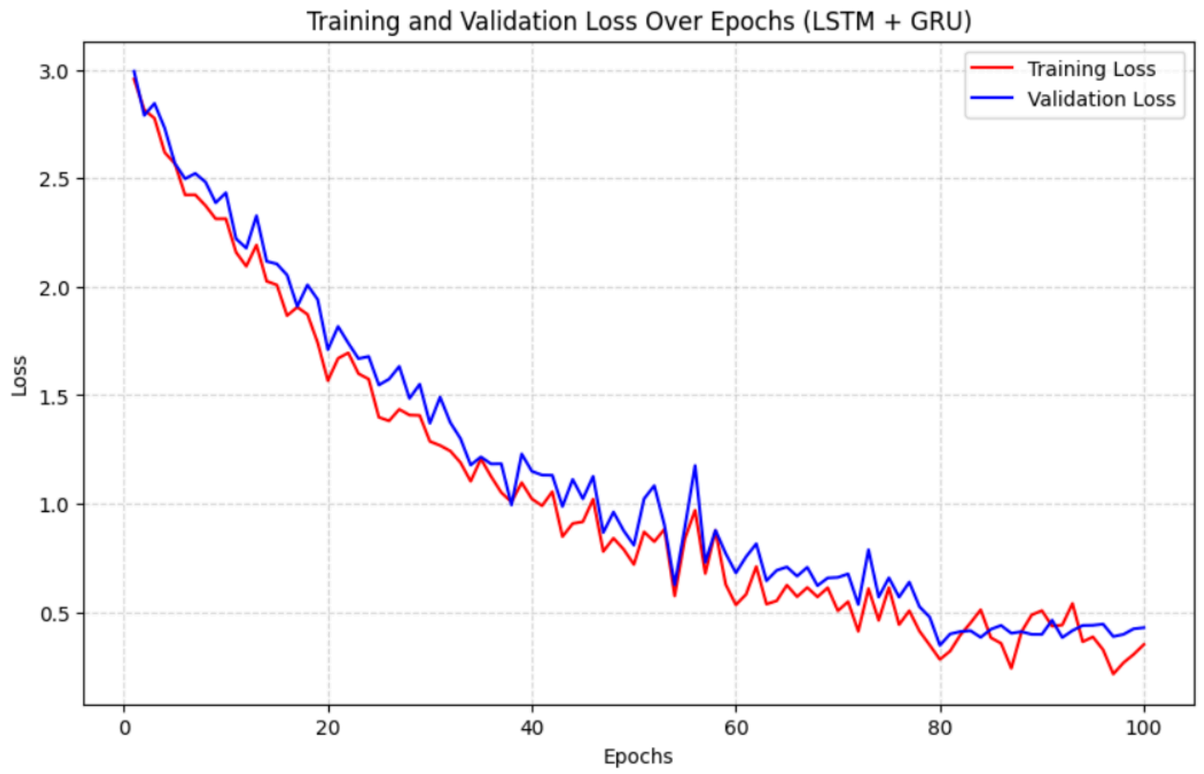
2. BIDIRECTIONAL LSTM + GRU ARCHITECTURE



RESULTS...

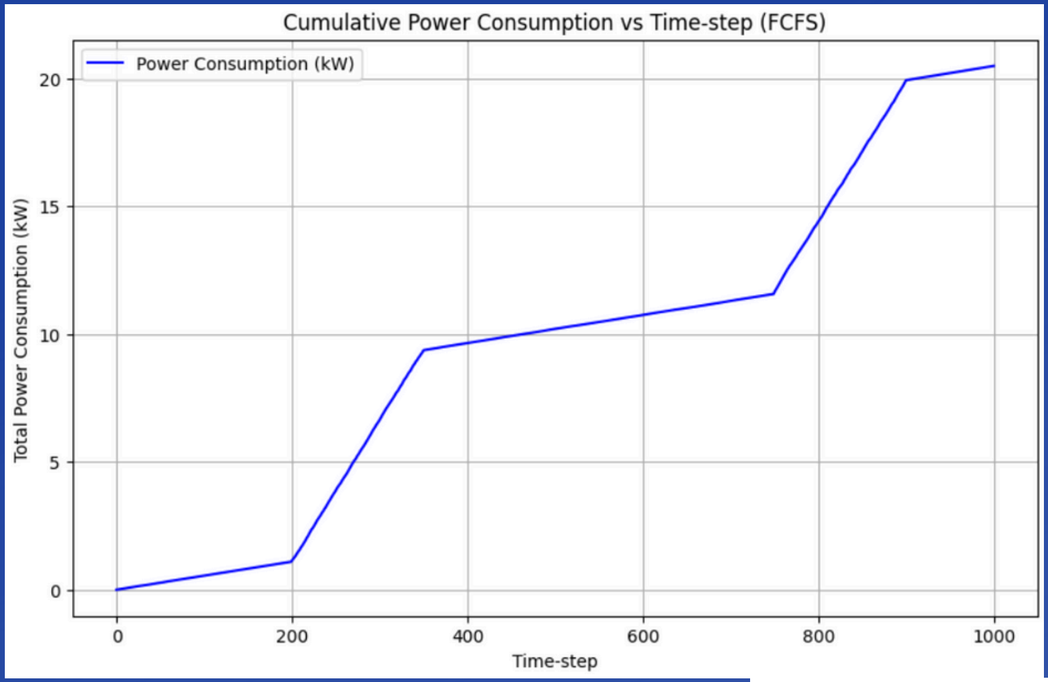
TRAINING HISTORY OF THE MODELS





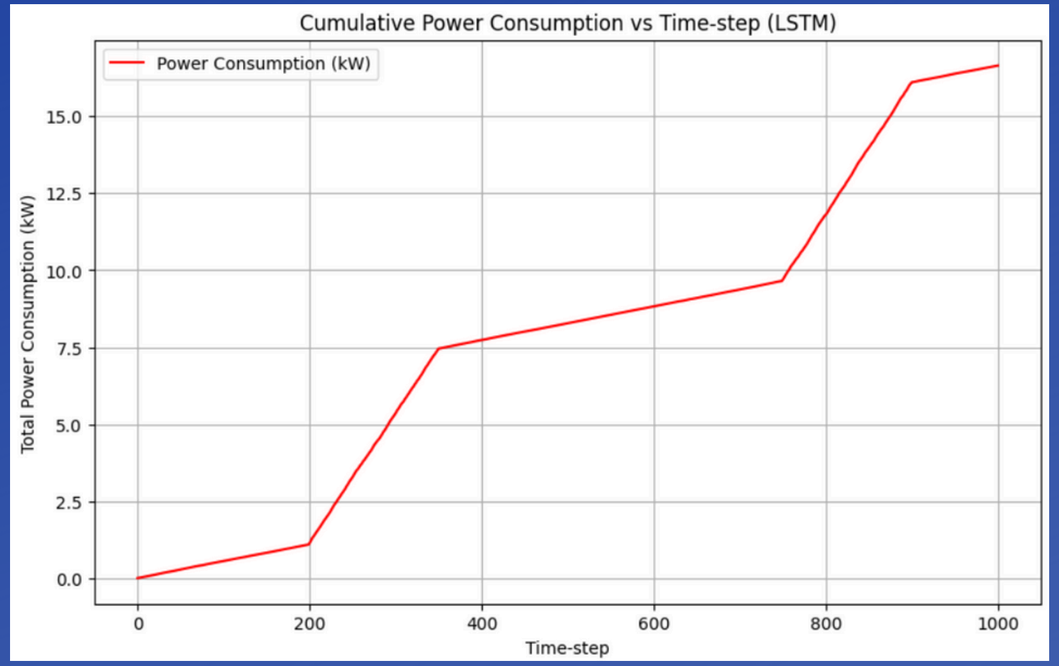
- Overall loss-
  - LSTM: Around 1.4
  - Bi-LSTM + GRU: Around 0.5

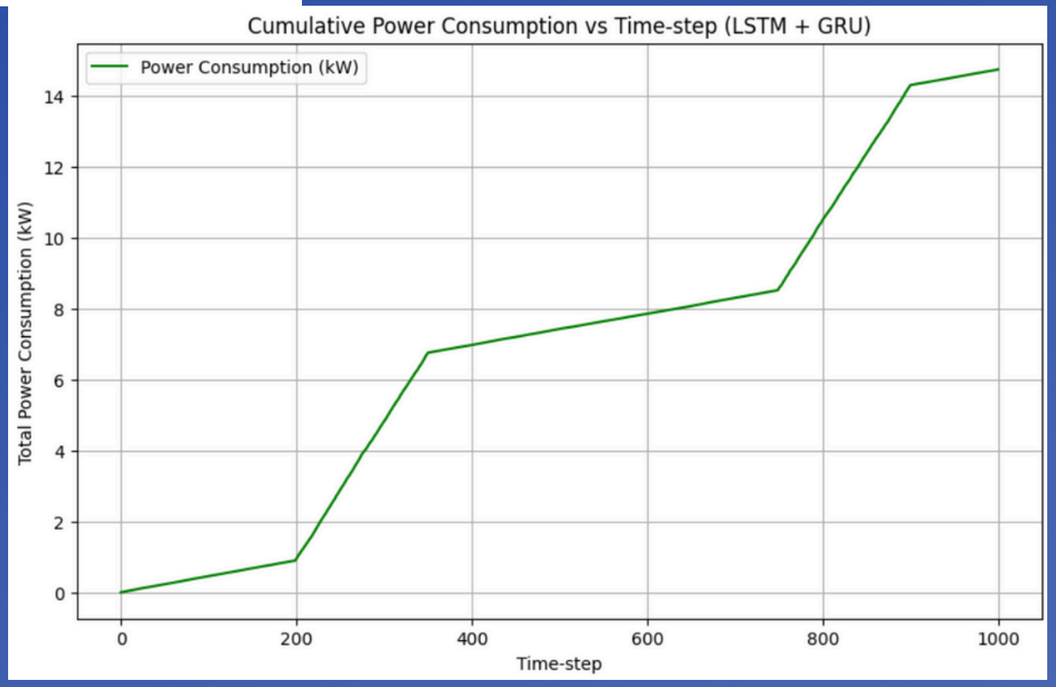
CUMULATIVE POWER CONSUMPTION EVOLUTION



FCFS
(20 KW)

LSTM +
GRU
(14 KW)





LSTM
(16 KW)

TO CONCLUDE...

- Machine Learning in multi-lift system seems to be comparatively more beneficial than in single-lift system
- Proposed architectures seem to show significant reductions in overall power consumption
- Active training might be able to overpower the mentioned pre-trained architectures

FURTHER WORK

- To implement a visual interface for the simulation software (for visual proof of advantage of ML-based lift systems)
- Further exploration of varied architectural models in search of improvements
- Testing with Real data
- Attempt at building live-training models

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THANK YOU